

## EAST POINT, PE, Canada

WMO: 714120

Lat: 46.4597N

Lon: 61.9878W

Elev: 8

StdP: 101.23

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -17.4      | -15.1 | -21.6                         | 0.5 | -16.7 | -19.0 | 0.7 | -14.4 | 19.7                  | -6.0 | 18.2 | -6.0 | 6.1                   | 250  | 0.776 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.8                    | 24.5            | 20.6 | 23.2 | 19.8 | 22.0 | 19.1 | 21.4                | 23.5 | 20.5 | 22.5 | 19.7 | 21.5 | 5.0                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.6                            | 15.3 | 22.8 | 19.7 | 14.5 | 21.8 | 18.9 | 13.8 | 21.0 | 62.4          | 23.6 | 59.2 | 22.6 | 56.5 | 21.6 | 27.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 17.6          | 15.3 | 13.2 | DB                         | -20.6 | 28.2               | 2.7 | 1.7                                                | -22.5 | 29.4       | -24.1 | 30.4       | -25.6 | 31.4       | -27.5 | 32.6 | (4) |
| (5)               |      |      | WB                         | -20.9 | 22.8               | 2.6 | 1.6                                                | -22.7 | 24.0       | -24.3 | 24.9       | -25.8 | 25.8       | -27.7 | 27.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 6.2         | -5.0  | -5.6  | -2.6  | 1.9   | 7.2   | 12.8  | 18.1  | 18.7  | 15.3  | 9.7   | 4.5   | -0.7  | (6)  |  |
|                                                                        | DBStd       | 9.15        | 5.08  | 4.77  | 3.96  | 2.99  | 3.23  | 3.32  | 2.33  | 2.34  | 2.91  | 3.37  | 3.84  | 4.26  | (7)  |  |
|                                                                        | HDD10.0     | 2200        | 467   | 438   | 391   | 243   | 101   | 11    | 0     | 0     | 1     | 48    | 169   | 332   | (8)  |  |
|                                                                        | HDD18.3     | 4482        | 725   | 671   | 649   | 492   | 346   | 168   | 33    | 24    | 98    | 269   | 416   | 591   | (9)  |  |
|                                                                        | CDD10.0     | 829         | 0     | 0     | 0     | 1     | 13    | 94    | 252   | 269   | 160   | 37    | 3     | 0     | (10) |  |
|                                                                        | CDD18.3     | 70          | 0     | 0     | 0     | 0     | 0     | 2     | 27    | 34    | 7     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 104         | 0     | 0     | 0     | 0     | 1     | 6     | 38    | 49    | 10    | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 7           | 0     | 0     | 0     | 0     | 0     | 1     | 2     | 3     | 1     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 6.4         | 7.3   | 6.8   | 6.6   | 6.3   | 5.7   | 5.5   | 5.1   | 5.2   | 6.1   | 7.1   | 7.3   | 7.9   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1069        | 88    | 75    | 79    | 80    | 75    | 83    | 73    | 97    | 106   | 117   | 114   | 114   | (15) |  |
|                                                                        | PrecMax     | 1348        | 246   | 208   | 181   | 141   | 131   | 218   | 123   | 209   | 334   | 197   | 240   | 276   | (16) |  |
|                                                                        | PrecMin     | 740         | 32    | 25    | 31    | 45    | 39    | 36    | 31    | 22    | 37    | 44    | 62    | 46    | (17) |  |
|                                                                        | PrecStd     | 162         | 46    | 41    | 37    | 28    | 25    | 40    | 29    | 50    | 64    | 44    | 40    | 46    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 7.7   | 6.0   | 7.1   | 13.4  | 20.1  | 23.9  | 26.4  | 26.7  | 24.6  | 20.0  | 15.1  | 10.6  | (19) |  |
|                                                                        |             | MCWB        | 7.1   | 5.1   | 5.8   | 10.6  | 14.8  | 18.6  | 21.7  | 21.9  | 20.6  | 17.0  | 14.0  | 9.9   | (20) |  |
|                                                                        | 2%          | DB          | 5.0   | 3.7   | 4.7   | 9.5   | 16.1  | 21.1  | 24.2  | 24.6  | 22.2  | 17.5  | 13.1  | 8.3   | (21) |  |
|                                                                        |             | MCWB        | 4.4   | 3.0   | 3.7   | 7.5   | 12.4  | 17.5  | 20.6  | 20.8  | 19.3  | 15.6  | 12.2  | 7.6   | (22) |  |
|                                                                        | 5%          | DB          | 3.2   | 2.2   | 3.3   | 7.3   | 13.4  | 19.1  | 22.9  | 23.4  | 20.8  | 16.0  | 11.7  | 6.9   | (23) |  |
|                                                                        |             | MCWB        | 2.5   | 1.5   | 2.3   | 5.9   | 10.6  | 16.4  | 19.8  | 20.0  | 18.3  | 14.3  | 10.8  | 6.2   | (24) |  |
|                                                                        | 10%         | DB          | 1.7   | 0.7   | 2.1   | 5.7   | 11.5  | 17.5  | 21.7  | 22.3  | 19.5  | 14.6  | 10.2  | 5.1   | (25) |  |
|                                                                        |             | MCWB        | 1.1   | 0.1   | 1.2   | 4.4   | 9.4   | 15.2  | 19.0  | 19.4  | 17.4  | 13.1  | 9.2   | 4.3   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 7.4   | 5.4   | 6.2   | 11.3  | 16.1  | 19.9  | 22.6  | 22.7  | 21.5  | 17.7  | 14.2  | 10.1  | (27) |  |
|                                                                        |             | MCDB        | 7.7   | 5.8   | 6.7   | 12.9  | 18.8  | 22.1  | 25.3  | 25.3  | 23.3  | 19.0  | 14.9  | 10.6  | (28) |  |
|                                                                        | 2%          | WB          | 4.6   | 3.2   | 4.0   | 8.0   | 12.9  | 18.2  | 21.4  | 21.6  | 20.0  | 15.9  | 12.5  | 7.7   | (29) |  |
|                                                                        |             | MCDB        | 4.9   | 3.5   | 4.4   | 9.0   | 15.2  | 20.1  | 23.5  | 23.7  | 21.8  | 17.0  | 12.9  | 8.1   | (30) |  |
|                                                                        | 5%          | WB          | 2.6   | 1.6   | 2.5   | 6.1   | 11.0  | 16.7  | 20.4  | 20.7  | 18.8  | 14.6  | 11.0  | 6.3   | (31) |  |
|                                                                        |             | MCDB        | 3.1   | 2.1   | 3.1   | 7.1   | 12.6  | 18.7  | 22.2  | 22.7  | 20.1  | 15.8  | 11.7  | 6.8   | (32) |  |
|                                                                        | 10%         | WB          | 1.1   | 0.1   | 1.2   | 4.6   | 9.7   | 15.4  | 19.6  | 19.9  | 17.8  | 13.3  | 9.3   | 4.4   | (33) |  |
|                                                                        |             | MCDB        | 1.6   | 0.6   | 2.0   | 5.5   | 11.4  | 17.3  | 21.2  | 21.9  | 19.2  | 14.4  | 10.1  | 5.1   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 6.6   | 6.3   | 5.3   | 5.0   | 5.9   | 5.8   | 5.7   | 5.8   | 5.6   | 5.1   | 5.0   | 5.3   | (35) |  |
|                                                                        |             | MCDBR       | 8.3   | 7.7   | 7.0   | 8.2   | 10.0  | 8.8   | 7.3   | 7.2   | 7.1   | 6.6   | 6.7   | 7.3   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 8.6   | 7.6   | 6.5   | 6.6   | 6.9   | 5.7   | 4.6   | 4.7   | 5.0   | 5.7   | 6.9   | 7.6   | (37) |  |
|                                                                        |             | MCWBR       | 8.3   | 7.7   | 6.9   | 7.7   | 9.0   | 7.9   | 6.6   | 6.3   | 6.1   | 6.2   | 6.9   | 7.4   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.301 | 0.293 | 0.324 | 0.351 | 0.371 | 0.383 | 0.411 | 0.399 | 0.360 | 0.345 | 0.326 | 0.303 | (40) |  |
|                                                                        |             | taud        | 2.387 | 2.293 | 2.281 | 2.400 | 2.430 | 2.412 | 2.383 | 2.418 | 2.504 | 2.502 | 2.466 | 2.444 | (41) |  |
|                                                                        | Ebn at Noon |             | 793   | 878   | 900   | 906   | 895   | 883   | 851   | 848   | 857   | 815   | 759   | 751   | (42) |  |
|                                                                        |             | Edh at Noon | 72    | 95    | 112   | 110   | 111   | 114   | 116   | 107   | 90    | 78    | 67    | 62    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 0.99  | 1.72  | 3.03  | 4.18  | 5.10  | 5.53  | 5.59  | 4.92  | 3.74  | 2.29  | 1.17  | 0.76  | (44) |  |
|                                                                        | RadStd      |             | 0.07  | 0.25  | 0.32  | 0.37  | 0.44  | 0.45  | 0.37  | 0.31  | 0.32  | 0.23  | 0.14  | 0.07  | (45) |  |

## Historical Trends

|                         |       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         | (d)   | (e)   | (f)     | (g)    | (h)     | (i)   | (j)   | (k)         | (l)     | (m)     |         |
| Station Only            | +0.59 | N/A   | +1.93   | N/A    | N/A     | N/A   | N/A   | N/A         | +71     | N/A     |         |
| Regional (27 neighbors) | N/A   | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     |         |

Nomenclature: See separate page